ORIGINAL ARTICLE · SYSTEMS PHYSIOLOGY

Pregnancy as a Model of Energetically Constrained Biological Growth

Metabolic Reserve, Stable Energy Flux, and Infrastructure Cost in Fetal Development

AUTHOR
Yuri Dmitrochenko

FIELD
Systems Physiology

TYPE
Theoretical Study

ABSTRACT

Pregnancy represents one of the most successful physiological examples of sustained rapid biological growth in multicellular organisms. Despite the extreme anabolic demands of fetal development, total maternal energy expenditure near term increases only moderately — by approximately 15–20% in singleton pregnancy and approximately 30–40% in twin pregnancy.

This observation suggests that biological growth is constrained less by maximal energy availability than by the capacity of the organism to maintain stable and continuous energy transport within vascular, metabolic, and regulatory limits. We propose that pregnancy provides a natural physiological model in which transport architecture, substrate delivery, vascular adaptation, and long-term metabolic stability prove more important than absolute energetic reserve.

01 · INTRODUCTION

The Physiological Paradox of Pregnancy

Human fetal development is characterized by extremely rapid biomass accumulation. Near the end of gestation, fetal mass may increase by approximately 25–35 grams per day while maintaining high oxygen and glucose consumption. Nevertheless, the total increase in maternal energy expenditure remains relatively modest.

EXPECTED

A rapidly growing biological system producing several kilograms of new tissue should require a proportional increase in whole-organism energy expenditure.

OBSERVED

Total maternal energy expenditure rises by only 15–20% in singleton pregnancy and 30–40% in twin pregnancy — substantially below the organism's maximum energetic capacity.

This creates an important physiological paradox pointing to the conclusion that growth limitation depends less on absolute energy production and more on the stability of energy transport, vascular architecture, substrate delivery, and the continuity of metabolic support.

This work examines fetal development as a systems-level energetic process with three interacting components: the fetus itself, the placental transport interface, and the maternal support infrastructure.

02 · FETAL GROWTH DYNAMICS

Physiological Parameters Across Gestation

Approximate physiological parameters throughout gestation reveal a progressive increase in all indices of metabolic activity and growth rate — with a characteristic deceleration toward the very end of pregnancy.

Table 1 · Fetal Physiological Parameters by Gestational Week

Week	Mass (g)	Growth (g/day)	O2 total (ml/min)	O2 per kg (ml/min/kg)	Glucose (mg/kg/min)
12	14	1	0.07	5.0	4.0
16	100	4	0.6	6.4	4.5
20	300	9	2.2	7.2	5.0
24	600	14	4.8	8.0	5.5
28	1,005	19	8.8	8.8	6.0
32	1,702	29	13.7	8.1	6.0
36	2,622	34	19.2	7.3	5.6
40	3,462	25	22.8	6.6	5.2

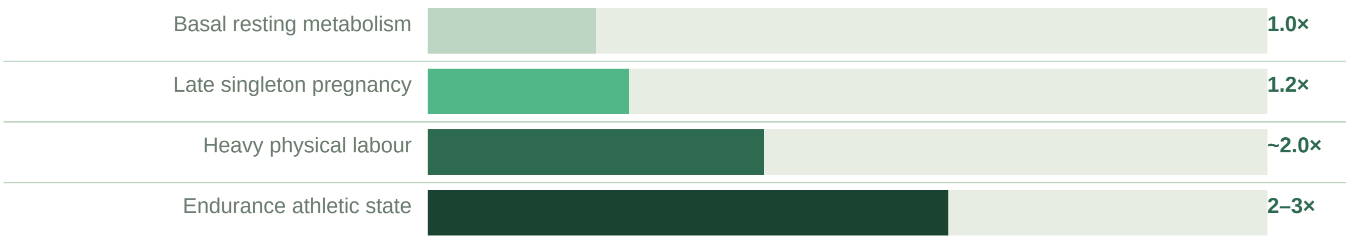
Noteworthy is the peak growth rate at week 36 (34 g/day), followed by a decline toward week 40. This late-gestation deceleration, despite preserved genetic programming, is one of the key indicators of resource-limited rather than program-limited growth.

03 · MATERNAL ENERGY RESERVE

Pregnancy and the Organism's Maximum Capacity

The healthy adult organism possesses metabolic reserve far exceeding the energetic demands of pregnancy. Comparing different metabolic states makes this gap immediately apparent.

Relative energy expenditure levels across physiological states



Professional endurance athletes and individuals performing prolonged heavy labour may sustain energy expenditures of 5,000–7,000 kcal per day and beyond. Thus, pregnancy itself does not approach the maximum energetic capacity of the human organism.

> *Physiological limitation arises primarily from transport continuity, vascular adaptation, placental exchange, microcirculatory stability, and long-term maintenance of metabolic flux — not from any deficit in total energy supply.*

04 · INFRASTRUCTURE COST OF GROWTH

The Price of the Supporting System

Near the end of pregnancy, the total maternal metabolic increase of approximately 20% can be conceptually divided into two roughly equal components.

~9%	~10%	~20%
Direct fetal metabolism	Supporting infrastructure	Total maternal metabolic increase

The supporting infrastructure encompasses the placenta, increased cardiac output, respiratory work, uterine growth, expanded blood volume, renal filtration, endocrine activity, and nutrient transport throughout the body.

The energetic cost of maintaining the growth-enabling system thus becomes comparable to the energetic cost of the growing tissue itself. This mirrors behaviour observed in large engineered systems, where infrastructure costs scale alongside productive capacity.

05 · MULTIPLE PREGNANCY

A Natural Model of Energetic Limitation

Twin and higher-order pregnancies provide an especially informative physiological model. As fetal number increases, total fetal mass rises — yet average mass per fetus decreases in a predictable pattern.

This demonstrates that biological growth does not scale linearly with available energetic reserve. Instead, the system encounters limitations in uterine perfusion, placental transport, oxygen delivery, glucose transfer, and spatial vascular architecture.

> *Near these limits, growth transitions from program-limited to resource-limited. This transition is especially visible during late gestation, when fetal growth velocity slows despite fully preserved developmental programming.*

06 · STABLE FLUX VS. MAXIMUM POWER

Two Fundamentally Different Metabolic Regimes

Physical exercise and pregnancy represent fundamentally different metabolic regimes, and conflating them leads to a misunderstanding of what actually limits fetal growth.

ATHLETIC EXERTION

Tolerates fluctuations, intermittent oxygen debt, temporary anaerobic metabolism, and recovery periods. Peak power output is the critical variable.

PREGNANCY

Requires uninterrupted perfusion, stable oxygen delivery, continuous substrate transport, and long-term metabolic stability sustained over months — with no capacity for recovery periods.

The fetoplacental system does not tolerate interruptions. Pregnancy therefore demonstrates that biological growth depends not on peak energy output, but on sustained, stable flux within transport constraints — a principle with broad implications well beyond obstetrics.

07 · CONCEPTUAL IMPLICATIONS

Relevance Beyond Pregnancy

Pregnancy may represent one of the clearest physiological demonstrations that biological growth is constrained less by maximal energy production than by the ability of a system to maintain stable and continuous energy distribution within transport and regulatory limits.

This conceptual framework may carry implications across several adjacent fields:

ONCOLOGY

Tumour energetics and transport constraints on proliferative growth

REGENERATIVE MEDICINE

Vascularisation principles for engineered tissue constructs

PLACENTAL PATHOLOGY

Fetal growth restriction as a failure of transport stability

The model also applies to the biology of ageing and to systems-level models of growth limitation wherever transport architecture — rather than total energetic reserve — sets the ceiling on anabolic activity.

08 · CONCLUSION

Pregnancy as a Physiological Model

Late pregnancy produces one of the highest sustained anabolic states in normal human physiology — yet requires only a moderate increase in whole-body maternal metabolism.

This indicates that biological growth efficiency is extremely high, that infrastructure cost constitutes a major share of total growth expenditure, and that transport architecture may be more limiting than total energetic reserve.

Multiple pregnancy further demonstrates the existence of systemic energetic and vascular limits that constrain growth even in the presence of preserved developmental programming. Pregnancy thus provides a compelling natural physiological model for studying energetically constrained growth in complex biological systems.

REFERENCES

1 Fowden A.L., Forhead A.J., Coan P.M., Burton G.J. The placenta and intrauterine programming. Journal of Neuroendocrinology. 2008.

2 Battaglia F.C., Meschia G. Fetal nutrition. Annual Review of Nutrition. 1988.

3 Hay W.W. Placental-fetal glucose exchange and fetal glucose metabolism. Transactions of the American Clinical and Climatological Association. 2006.

4 Carter A.M. Placental oxygen consumption. Placenta. 2000.

5 Blackburn S.T. Maternal, Fetal, and Neonatal Physiology. Elsevier.

6 Hytten F.E., Leitch I. The Physiology of Human Pregnancy. Blackwell Scientific Publications.

7 Durnin J.V.G.A. Energy requirements of pregnancy. Proceedings of the Nutrition Society. 1987.

8 Speakman J.R., Krol E. Maximal sustained energy budgets in humans and animals. Nature Communications. 2010.

9 Pontzer H. et al. Constrained Total Energy Expenditure and the Evolutionary Biology of Energy Balance. Exercise and Sport Sciences Reviews. 2016.

10 Burton G.J., Fowden A.L. The placenta: a multifaceted, transient organ. Philosophical Transactions of the Royal Society B. 2015.